

CUMMULATIVE HIERARCHIES OF UNIVERSES AND THEIR EQUIVALENCE IN TYPE THEORY

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University of Cambridge, SANDWICH Seminar (13/04/2026)

1. Introduction

INTRODUCTION : MOTIVATION

- *Martin L f Type Theory (MLTT)* :

$$A, B, a, b ::= \Pi A B \mid \lambda_{x:A} b \mid c a$$

- Universes: Manipulate types as terms, embed them in datatypes, pass them to functions...
- For instance, universe are necessary to *state* univalence

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- First idea : We introduce the type of all types U !

$$A, B, a, b ::= \Pi A B \mid \textcolor{red}{U} \mid \lambda_{x:A} b \mid c a$$

- Now types are also terms, for instance: $\Pi A B : U$
 - But... where does U itself live ?
- The only choice is to require $U : U$.

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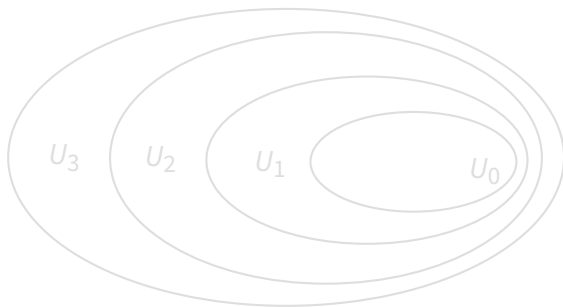
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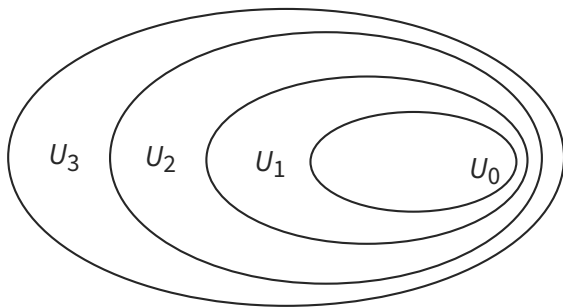
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- We need a universe hierarchy:



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$$A, B, a, b ::= \Pi A B \mid U_n \mid \lambda_{x:A} b \mid c a$$

- Russell Universes $\Rightarrow$ types and terms are conflated

$$U_0 : U_1, \quad U_1 : U_2, \quad \dots \quad U_n : U_{n+1}, \quad \dots$$

$$\frac{\Gamma \vdash A : U_n}{\Gamma \vdash A \text{ Type}}$$

- We can ask those universes to be **cumulative** :

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- Universes **à la Tarski** : A *fully explicit* presentation

$$A, B ::= \Pi A B \mid U_n \mid \text{El}_n(a)$$

$$a, b ::= \lambda_{x:A} b \mid c a \mid \Pi^n a b \mid \uparrow_n^m(a) \mid U_n^m$$

- With the rules :

$$\frac{\Gamma \vdash a : U_n}{\Gamma \vdash \text{El}_n(a) \text{ type}}$$

$$\frac{\Gamma \vdash a : U_n}{\Gamma \vdash \uparrow_n^m(a) : U_m}$$

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- Universes à la Russell $\Rightarrow$ Often considered as an informal presentation

$\rightarrow$ Relating the two: theories presented à la Russell have rigorous interpretation in the usual models of type theory

$\rightarrow$ Extension of initiality

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→ Extension of initiality

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- According to Luo : Russel $\nleftrightarrow$ Tarski with *structural sub-typing*
- Recent work by Assaf, Thir   : Equivalence proof for the calculus of construction, with only β as a reduction, doesn't adapt to η -equivalence.
- Cumulativity is still not widely adopted in proof assistants
 - $\Rightarrow$ Rocq : Cumulative universes with sub-typing judgement $A \leq B$
 - $\Rightarrow$ Agda, Lean : Non-cumulative universes

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- ⇒ Show syntactic equivalence of the two presentations for cumulative universes
- ⇒ Relate generalised algebraic theory (GAT) and pure type system (PTS) presentations of type theory.

To do this:

- Key lemmas : injectivity of type constructors and "no-confusion", which we can prove as a corollary of normalisation.
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2. GATs and Initiality

GENERALISED ALGEBRAIC THEORIES

- GATs : dependent type theory in the syntactic sense, presented in an *algebraic* way.
- Dependent counterpart of algebraic theories
- Gives a natural notion of semantics in Categories with Families : describe a type theory by expliciting its GAT of CwFs
- Everything is explicit and annotated:

$$\Gamma : \text{cxt}, A : \text{Ty}(\Gamma), B : \text{Ty}(\Gamma.A) \vdash \Pi(A, B) : \text{Ty}(\Gamma)$$

$$\Gamma : \text{cxt} A : \text{Ty}(\Gamma), B : \text{Ty}(\Gamma.A), b : \text{Tm}(\Gamma, B) \vdash \lambda(A, B, b) : \text{Tm}(\Gamma, \Pi(A, B))$$

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GAT OF TYPE THEORY WITH A UNIVERSE HIERARCHY

- GAT of CwFs with extra structure : Four sorts (Contexts, Substitutions, Types, Terms), operations describing the substitution calculus and logical structure

→ Extend with a cumulative universe hierarchy à la Tarski (Theory $\mathcal{T}_G$):

$$\Gamma : \text{cxt}, a : \text{Tm}(\Gamma, (U_l)_\Gamma), b : \text{Tm}(\Gamma.\text{El}_l(a), (U_l)_\Gamma) \vdash \Pi^l(a, b) : \text{Tm}(\Gamma, (U_l)_\Gamma)$$

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EQUATIONS

We require the following equations:

$$\text{El}_m((U_l^m)_\Gamma) = (U_l)_\Gamma$$

$$\text{El}_l(\Pi^l(a, b)) = \Pi(\text{El}_l(a), \text{El}_l(b))$$

$$\text{El}_m(\uparrow_l^m(a)) = \text{El}_l(a)$$

$$\uparrow_m^n \uparrow_l^m(a) = \uparrow_l^n(a)$$

$$\uparrow_m^n((U_l^m)_\Gamma) = (U_l^n)_\Gamma$$

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INITIALITY

- Voevodsky:

"A crucial component of this approach is the expected result that for a particular class of inference rules the term model is an initial object in the category of models. This is known as the Initiality Conjecture. In the case of the pure Calculus of Constructions with a “decorated” application operation this conjecture was proved in 1988 by Thomas Streicher [32]. The problem of finding an appropriate formulation of the general version of the conjecture and of proving this general version will be the subject of future work."

- Relies on subtle details in the formulation of the system.
- For $\mathcal{T}_G$, the recent work "*A Generalized Algebraic Theory for Type Theory with Explicit Universe Polymorphism*" (Bezem, Coquand, Dybjer, Escardo 2026) present a proof of initiality

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3. Other presentations of type theory

IMPLICIT SUBSTITUTIONS

- Make substitution *implicit*: externalise it as a meta-theoretical operation

→ Defines a theory $\mathcal{T}_T$, which we now express using inference rules:

$$\frac{\Gamma \vdash a : U_n \quad \Gamma, \text{El}_n(a) \vdash b : U_n}{\text{El}_n(\Pi^n a b) = \Pi(\text{El}_n(a), \text{El}_n(b))}$$

$$\frac{\Gamma \vdash a : U_n \quad \Gamma, \text{El}_n(a) \vdash b : U_n}{\uparrow_n^p(\Pi^n a b) = \Pi^p(\uparrow_n^p(a), \uparrow_n^p(b))} (n < p)$$

$$\frac{}{\Gamma \vdash \text{El}_n(U_m^n) = U_m}$$

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RUSSELL UNIVERSSES

- Now, the theory with *Russell* universes $\mathcal{T}_R$:

$$\frac{\Gamma \vdash A : U_n}{\Gamma \vdash A \text{ Type}}$$

$$\frac{}{\Gamma \vdash U_n : U_p} (n < p)$$

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$$\frac{\Gamma \vdash A : U_n \quad \Gamma.A \vdash B : U_n}{\Gamma \vdash \Pi(A, B) : U_n}$$

TARSKI TO RUSSELL

We define an erasure operation $|\cdot|$ from $\mathcal{T}_T$ to $\mathcal{T}_R$ that removes codes, liftings and decodings.

- $|U_n| = U_n$.
- $|U_m^n| = U_m$.
- $|\text{app}(A, B, c, a)| = \text{app}(|A|, |B|, |c|, |a|)$.
- $|\lambda(A, B, b)| = \lambda(|A|, |B|, |b|)$.
- $|v_i| = v_i$.
- $|\text{El}_n(a)| = |a|$.
- $|t_m^n(a)| = |a|$.
- $|\Pi^n a b| = \Pi(|a|, |b|)$.
- $|\Pi(A, B)| = \Pi(|A|, |B|)$.

PURE TYPE SYSTEM : PRESENTATION WITHOUT ANNOTATIONS

- Our last theory $\mathcal{T}_P$ get rids of useless annotations in app and λ to get a formulation of the theory as a pure type system:

$\lambda(A, B, b) \mid \text{app}(A, B, c, a)$

(System $\mathcal{T}_R$)

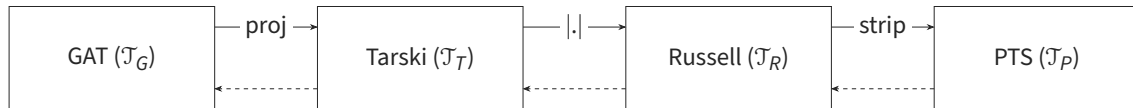
$\xRightarrow{\text{strip}}$

$\lambda_A b \mid c a$

(System $\mathcal{T}_P$)

SUMMARY

Those four theories have forgetful / erasure maps:



$\Rightarrow$ Goal find inverses : any judgement can be lifted back to a more explicit theory in a unique way.

4. Normalisation

NORMALISATION

- We want to show normalisation for $\mathcal{T}_G$
 - Use Artin Gluing, which constructs a normalisation model of our theory
- Proof relevant version of the usual proof by reducibility candidates.
- We need to consider open terms, thus we work internally to the presheaf topos.

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NORMALISATION MODEL

- Define neutral terms $\text{Neut}(\Gamma)$ and normal form $\text{Type}_N(\Gamma)$ and $\text{Elem}_N(\Gamma, T)$.
- Here are the neutral and normal forms for the universes construction :

$\text{El}_n k \in \text{Type}_N(X)$ if $k \in \text{Neut}(X, U_n)$.

$\Pi^n k l \in \text{Elem}_N(X, U_n)$ if $k \in \text{Elem}_N(X, U_n)$ and $l \in \text{Elem}_N(X, \langle \text{El}_n k \rangle, U_n)$.

$U_m^n \in \text{Elem}_N(X, U_n)$.

$\uparrow_m^n k \in \text{Elem}_N(X, U_n)$ if $k \in \text{Neut}(X, U_m)$.

$k \in \text{Elem}_N(X, U_n)$ and $t \in \text{Elem}_N(X, \langle \text{El}_n k \rangle)$, if $k \in \text{Neut}(X, U_n)$ and $t \in \text{Neut}(X, \langle \text{El}_n k \rangle)$.

NORMALISATION MODEL : TARSKI-STYLE UNIVERSES

- Construct the normalisation model inductively, equip terms with a reducibility predicate.
 - The main idea : Being reducible at a universe is the same thing as being reducible at the decoded type.
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CONSEQUENCES

Finally, we obtain normalisation:

$$\text{nf}_{\text{Ty}} : \text{Type}_{M_0}(\Gamma) \rightarrow \text{Type}_N(\Gamma)$$

and

$$\text{nf}_{\text{Tm}} : \text{Elem}_{M_0}(\Gamma, A) \rightarrow \text{Elem}_N(\Gamma, A)$$

Injectivity of type constructors in the syntactic model M_0 :

$$\prod A B = \prod A' B' \implies A = A' \text{ and } B = B'$$

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5. Equivalence results

EQUIVALENCE BETWEEN $\mathcal{T}_G$ AND $\mathcal{T}_T$

Briefly, this is the equivalence between *explicit* and *implicit* substitutions.

$$\text{proj}(\Gamma \vdash J) = \text{proj}(\Gamma) \vdash \text{proj}(J)$$

$$\text{proj}(\sigma : \Gamma \rightarrow \Delta) = \vdash \text{ (empty judgement)}$$

$$\text{proj}([a]) = [\text{proj}(a)]$$

$$\text{proj}(A\sigma) = (\dots(\text{proj}(A)\sigma_1)\dots)\dots\sigma_n$$

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$$\text{proj}(q) = v_0$$

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$$\text{proj}(\sigma\delta) = \text{proj}(\sigma) :: \text{proj}(\delta)$$

$$\text{proj}(\sigma^+) = (\sigma_1^+, \dots, \sigma_n^+)$$

→ No composition in the implicit case, output a list that is unfolded when applied

→ Correction result :

If $\Gamma \vdash J$ in $\mathcal{T}_G$ then $\text{proj}(\Gamma \vdash J)$ is derivable in $\mathcal{T}_T$.

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→ No composition in the implicit case, output a list that is unfolded when applied

→ Correction result :

If $\Gamma \vdash J$ in $\mathcal{T}_G$ then $\text{proj}(\Gamma \vdash J)$ is derivable in $\mathcal{T}_T$.

EQUIVALENCE BETWEEN $\mathcal{T}_G$ AND $\mathcal{T}_T$

Briefly, this is the equivalence between *explicit* and *implicit* substitutions.

$$\text{proj}(\Gamma \vdash J) = \text{proj}(\Gamma) \vdash \text{proj}(J)$$

$$\text{proj}(\sigma : \Gamma \rightarrow \Delta) = \vdash \text{ (empty judgement)}$$

$$\text{proj}([a]) = [\text{proj}(a)]$$

$$\text{proj}(A\sigma) = (\dots(\text{proj}(A)\sigma_1)\dots)\dots\sigma_n$$

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$$\text{proj}(q) = v_0$$

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EQUIVALENCE BETWEEN $\mathcal{T}_T$ AND $\mathcal{T}_R$: UNIQUENESS LEMMA

Now for $\mathcal{T}_T \Leftrightarrow \mathcal{T}_R$ (*Russell and Tarski universes*).

We would want to prove the following lemma :

- If $\Gamma \vdash A$ type and $\Gamma \vdash B$ type in $\mathcal{T}_T$, with $|A| = |B|$, then $\Gamma \vdash A = B$.
- If $\Gamma \vdash u_0 : A_0$ and $\Gamma \vdash u_1 : A_1$ in $\mathcal{T}_T$ with $|u_0| = |u_1|$ then : $\Gamma \vdash A_0 = A_1$ and $\Gamma \vdash u_0 = u_1 : A_0$.

i.e. Terms and types in $\mathcal{T}_T$ with the same erasure in $\mathcal{T}_R$ should be convertible !

→ Or not... take x and $\uparrow_n^m x$ for instance.

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$\Rightarrow$ Two judgments in $\mathcal{T}_T$ with the same erasure are convertible *modulo lifting* !

- Proof by mutual induction on the size of A and u_0 .
- Injectivity of $U_\bullet$ and no-confusion between $U_\bullet$ and $\Pi(-, -)$ are necessary.

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EQUIVALENCE BETWEEN $\mathcal{T}_T$ AND $\mathcal{T}_R$: UNIQUENESS LEMMA, AN EXAMPLE

If u_0 is a λ -abstraction, by inversion, the derivation $\Gamma \vdash u_0 : A_0$ looks like :

$$\frac{\frac{\Pi_0}{\Gamma, C_0 \vdash b_0 : B_0}}{\Gamma \vdash \lambda(C_0, B_0, b_0) : \Pi(C_0, B_0)}$$

Then $|u_0| = \lambda(|C_0|, |B_0|, |b_0|) = |u_1|$ and so u_1 is necessarily an abstraction :

$$u_1 = \lambda(C_1, B_1, b_1)$$

with $|C_1| = |C_0|, |B_1| = |B_0|, |b_1| = |b_0|$.

Then we reason by inversion and use the induction hypothesis to construct the derivation.

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EQUIVALENCE BETWEEN $\mathcal{T}_T$ AND $\mathcal{T}_R$: UNIQUENESS LEMMA, AN EXAMPLE

For each case we prove a "catch-up lemma", for instance:

- If $\Gamma \vdash \lambda(A, B, b) : A_0$ and $\Gamma \vdash t : A_1$ in $\mathcal{T}_T$ with $|\lambda(A, B, b)| = |t|$ then :
 - 1st case : $\Gamma \vdash A_0 = A_1$ and $\Gamma \vdash \lambda(A, B, b) = t : A_0$.
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- Proof by induction on t .
- Only interesting cases : $t = \lambda(A', B', b')$ or $t = \uparrow_n^m (t')$.

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We now want to show that we can uniquely lift judgements from $\mathcal{T}_R$ to $\mathcal{T}_T$:

If $\Gamma \vdash J$ in $\mathcal{T}_R$ then there exists a unique judgment $\Gamma_T \vdash J_T$ modulo conversion in system $\mathcal{T}_T$ such that $|\Gamma_T| = \Gamma$ and $|J_T| = J$.

Proof : by induction on the rules of $\mathcal{T}_R$. This exhibits a translation from $\mathcal{T}_R$ to $\mathcal{T}_T$.

We use the uniqueness lemma to "unify" the translations.

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If the last applied rule is :

$$\frac{\frac{\Pi_1}{\Gamma \vdash A \text{ type}} \quad \frac{\Pi_2}{\Gamma, A \vdash B \text{ type}}}{\Gamma \vdash \Pi(A, B) \text{ type}}$$

By the IH : $\Gamma' \vdash A' \text{ type}$ and $\Gamma'', A'' \vdash B' \text{ type}$ with $|\Gamma'| = |\Gamma''| = \Gamma$ and $|A''| = |A'| = A$.

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Now $\mathcal{T}_R \Leftrightarrow \mathcal{T}_P$: removing annotations.

Same proof scheme, but we can prove this directly:

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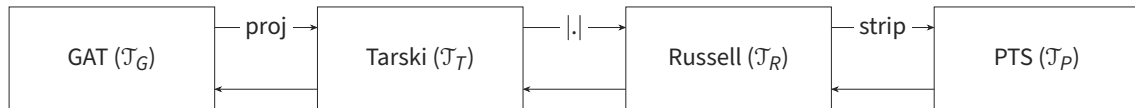
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SUMMARY



EXTENDING THE RESULT

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Thank you for your attention.